

Emotion-Aware Multimodal Sentiment Analysis Using Prototype Memory

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Abstract—Multimodal sentiment analysis uses information from different modalities, such as text, audio, and video, to understand human sentiment more effectively. Each modality provides different information, and its importance may vary depending on the input sample. In this work, textual, acoustic, and visual features are extracted using pretrained models and processed through a multimodal framework. The modalities are ranked based on their importance and progressively combined using cross-attention-based fusion. To further improve the fused representation, an Emotion Prototype Memory (EPM) module is introduced. The EPM module learns representative patterns for different sentiment classes and uses them to refine the fused multimodal features before final classification. The proposed approach is evaluated on the MELD dataset for three sentiment classes: positive, negative, and neutral. Experimental results show that the addition of EPM improves the performance of the multimodal sentiment classification model, demonstrating the usefulness of prototype-based refinement for sentiment prediction.

Index Terms—Multimodal Sentiment Analysis, Emotion Prototype Memory, Multimodal Fusion, Cross-Attention, Sentiment Classification

I. INTRODUCTION

Sentiment analysis is an important area of Natural Language Processing (NLP) that focuses on identifying opinions and sentiments expressed by people. Traditional sentiment analysis methods mainly use textual information to classify sentiment as positive, negative, or neutral. However, human sentiment is not expressed only through words. Voice tone, speaking style, and visual expressions can also provide useful information about a person's sentiment. Therefore, using multiple modalities can provide a better understanding of human sentiment.

Multimodal sentiment analysis combines information from different modalities, mainly text, audio, and video. Text provides information through spoken or written words, audio captures information related to speech and voice patterns, and video provides visual information. Combining these modalities can improve sentiment prediction because each modality provides different and complementary information.

One of the main challenges in multimodal sentiment analysis is that all modalities may not contribute equally to every input sample. In some cases, textual information may provide stronger sentiment information, while in other cases, audio or visual information may be more useful. Therefore, assigning the same importance to every modality may not always produce the best representation. Dynamic modality

importance can help address this problem by estimating the contribution of each modality before the fusion process.

Another challenge is effectively combining information from different modalities. Simple feature concatenation may not fully capture the relationships between text, audio, and video. Attention-based fusion methods allow the model to learn interactions between modalities and focus on useful information. In this work, modality features are ranked according to their importance and progressively combined using a cross-attention-based fusion process.

Although multimodal fusion provides a combined representation of different modalities, the final representation may still contain overlapping information between sentiment classes. To improve the discriminative ability of the fused representation, this work introduces an Emotion Prototype Memory (EPM) module. The EPM module learns representative prototype patterns related to different sentiment classes. The fused multimodal representation is compared with these learned prototypes, and the obtained prototype information is used to refine the representation before final sentiment classification.

The proposed framework is evaluated using the MELD dataset, which contains textual, acoustic, and visual information from multi-party conversations. Text features are extracted using BERT, audio features are obtained using HuBERT, and visual features are extracted using CLIP with a ViT-B/32 visual encoder. The extracted features are reduced and projected into a common representation space before modality importance estimation and multimodal fusion. Finally, the EPM module refines the fused features, and the resulting representation is used to predict positive, neutral, or negative sentiment.

The main contribution of this work is the integration of an Emotion Prototype Memory module into a multimodal sentiment analysis framework. The proposed EPM module introduces prototype-based refinement after multimodal fusion, allowing the model to use learned sentiment-related patterns before making the final prediction. Experimental results show that the addition of EPM improves the performance of the multimodal model, indicating that prototype-based refinement can provide useful information for multimodal sentiment classification.

II. RELATED WORK

A. Sentiment Analysis

Sentiment analysis is used to identify the sentiment expressed in textual content and generally classifies it as positive,

negative, or neutral. Earlier approaches mainly used traditional machine learning methods such as Naive Bayes, Support Vector Machines (SVM), and Logistic Regression. With the development of deep learning and transformer-based models, contextual language models such as BERT have improved text representation by capturing the meaning of words based on their context [1].

B. Multimodal Sentiment Analysis

Text alone may not always provide complete information about human sentiment. Audio can provide information through speech patterns and voice characteristics, while visual information can provide additional cues from video frames. Multimodal sentiment analysis combines these different sources of information to obtain a more complete representation of sentiment. The MELD dataset is commonly used for multimodal emotion and sentiment analysis and provides textual, acoustic, and visual information from conversations [2].

C. Multimodal Feature Representation

Pretrained models have been widely used to obtain meaningful representations from different modalities. BERT can be used to obtain contextual representations from textual information [1]. HuBERT learns speech representations from audio signals and can be used for acoustic feature extraction [3]. Similarly, CLIP learns relationships between visual and textual information and provides useful visual representations through its image encoder [4]. These pretrained models reduce the need to train separate feature extractors from the beginning.

D. Attention-Based Multimodal Fusion

Attention mechanisms have become important for combining information from different modalities. The Transformer architecture introduced self-attention to capture relationships between different elements of an input sequence [5]. In multimodal learning, cross-modal attention allows information from one modality to interact with information from another modality. Multimodal Transformer-based approaches have shown that cross-modal interactions can help learn useful relationships between different modalities [6].

E. Dynamic Modality Importance

Different modalities may not provide the same amount of useful information for every input. For example, text may be more informative for one sample, while audio or visual information may be more useful for another. Therefore, dynamically estimating the importance of each modality can help the model decide which information should receive more attention during fusion. In this work, modality importance is estimated before the fusion stage, and the modalities are ranked based on their estimated importance.

F. Prototype-Based Learning

Prototype-based learning represents different classes using learned representative patterns called prototypes. Instead of depending only on the final fused representation, a model can compare the input representation with learned prototypes to obtain additional class-related information. This approach can help the model learn more meaningful differences between classes.

Based on this idea, the proposed work introduces an Emotion Prototype Memory (EPM) module after multimodal fusion. The EPM module maintains learnable prototype representations and compares them with the fused multimodal features. The resulting prototype information is then used to refine the fused representation before final sentiment classification. This addition aims to improve the ability of the model to distinguish between positive, neutral, and negative sentiment classes.

III. PROPOSED METHODOLOGY

This section describes the proposed multimodal sentiment analysis framework. The framework processes three different modalities, namely text, audio, and video, to capture complementary information for sentiment prediction. Pretrained models are used to extract feature representations from each modality, followed by dimensionality reduction to obtain compact feature vectors. These features are projected into a common embedding space and refined using a Transformer-based context encoder. The encoded features are then ranked according to their importance through a Dynamic Modality Importance (DMI) module and combined using cascaded cross-attention fusion. To further improve the discriminative ability of the fused representation, an Emotion Prototype Memory (EPM) module is introduced. The refined feature representation is finally passed to a classifier to predict the sentiment class as positive, neutral, or negative. Figure 1 illustrates the overall architecture of the proposed framework.

A. Overall Architecture

The proposed framework consists of six major stages: multimodal feature extraction, feature dimensionality reduction, modality projection and context encoding, dynamic modality importance estimation, cascaded cross-attention fusion, and sentiment classification with Emotion Prototype Memory. The complete architecture is designed to effectively utilize complementary information from textual, acoustic, and visual modalities while reducing redundant or less informative features.

Initially, textual features are extracted using the BERT model, audio features are obtained using HuBERT, and visual features are extracted using the CLIP ViT-B/32 image encoder. Since the extracted feature vectors have different dimensions, Principal Component Analysis (PCA) is applied to obtain compact representations while preserving the most informative characteristics of each modality.

After dimensionality reduction, the feature vectors are projected into a common feature space using linear projection layers. A Transformer-based context encoder is then employed

to capture contextual relationships within the projected representations before multimodal fusion.

The encoded modality representations are passed to the Dynamic Modality Importance (DMI) module, which estimates the contribution of each modality for the current sample. Based on the estimated importance scores, the modalities are ranked and progressively fused through cascaded cross-attention. This strategy allows the framework to emphasize the most informative modality while still incorporating complementary information from the remaining modalities.

The fused multimodal representation is further refined using the proposed Emotion Prototype Memory (EPM) module. The EPM module stores learnable prototype representations corresponding to different sentiment patterns. By comparing the fused representation with these prototypes, the model obtains additional class-specific information that helps distinguish between similar sentiment categories.

Finally, the refined feature representation is passed through a fully connected classification layer followed by a Softmax activation function to predict one of the three sentiment classes: positive, neutral, or negative.

B. System Workflow

The overall workflow of the proposed framework is shown in Figure 2. The process begins by taking three input modalities, namely text, audio, and video, from the MELD dataset. Each modality is processed independently to extract meaningful feature representations using pretrained models.

The textual modality is processed using BERT to obtain contextual embeddings. Audio signals are processed using HuBERT to extract acoustic representations, while visual information is extracted from video frames using the CLIP ViT-B/32 image encoder. These pretrained models generate high-dimensional feature vectors that effectively capture semantic, acoustic, and visual information.

Since the extracted features have different dimensions, Principal Component Analysis (PCA) is applied to reduce the feature dimensions while preserving the most informative characteristics of each modality. The reduced features are then projected into a common embedding space using linear projection layers, enabling effective interaction between different modalities.

Next, the projected features are passed through a Transformer-based context encoder to capture contextual dependencies within each modality. The encoded representations are then provided to the Dynamic Modality Importance (DMI) module, which estimates the relative importance of text, audio, and video for the current sample. Based on these importance scores, the modalities are ranked before the fusion process.

The ranked modality representations are progressively combined using cascaded cross-attention fusion. Instead of directly concatenating all features, cross-attention enables the model to learn interactions between modalities and generate a richer multimodal representation.

To further improve the fused representation, the proposed Emotion Prototype Memory (EPM) module is introduced

after multimodal fusion. The EPM module stores learnable prototype representations corresponding to different sentiment patterns. The fused feature representation is compared with these prototypes, and the retrieved prototype information is used to refine the final multimodal representation.

Finally, the refined representation is passed through a fully connected classification layer followed by a Softmax activation function to predict one of the three sentiment classes: positive, neutral, or negative.

C. Multimodal Feature Extraction

The proposed framework utilizes three different modalities, namely text, audio, and video, to capture complementary information for sentiment analysis. Each modality is processed independently using a pretrained model that has demonstrated strong performance in its respective domain. The extracted feature representations are then used for subsequent dimensionality reduction and multimodal fusion.

1) *Text Feature Extraction*: Textual information carries the semantic content of an utterance and plays a significant role in sentiment analysis. In the proposed framework, textual features are extracted using the Bidirectional Encoder Representations from Transformers (BERT) model. BERT generates contextual embeddings by considering both the left and right context of each word, allowing it to capture the semantic meaning of the complete sentence.

Each utterance is first tokenized using the BERT tokenizer and then passed through the pretrained BERT encoder. The contextual representation corresponding to the special [CLS] token is used as the sentence-level representation since it summarizes the overall meaning of the input utterance. The extracted feature vector has a dimension of 768.

The textual feature representation can be expressed as

$$T = \text{BERT}(X_t), \quad (1)$$

where X_t represents the input text sequence and $T \in \mathbb{R}^{768}$ denotes the extracted textual feature vector.

2) *Audio Feature Extraction*: Audio provides useful emotional cues through speech characteristics such as tone, pitch, and speaking style. In the proposed framework, acoustic features are extracted using the HuBERT model. HuBERT is a self-supervised speech representation model that learns meaningful acoustic representations without requiring manually annotated labels during pretraining.

Each speech signal is first converted into the required input format and passed through the pretrained HuBERT encoder. The encoder generates a contextual feature representation that captures important acoustic characteristics of the utterance. A 768-dimensional feature vector is obtained for each audio sample.

The audio feature extraction process is represented as

$$A = \text{HuBERT}(X_a), \quad (2)$$

where X_a denotes the input speech signal and $A \in \mathbb{R}^{768}$ represents the extracted acoustic feature vector.

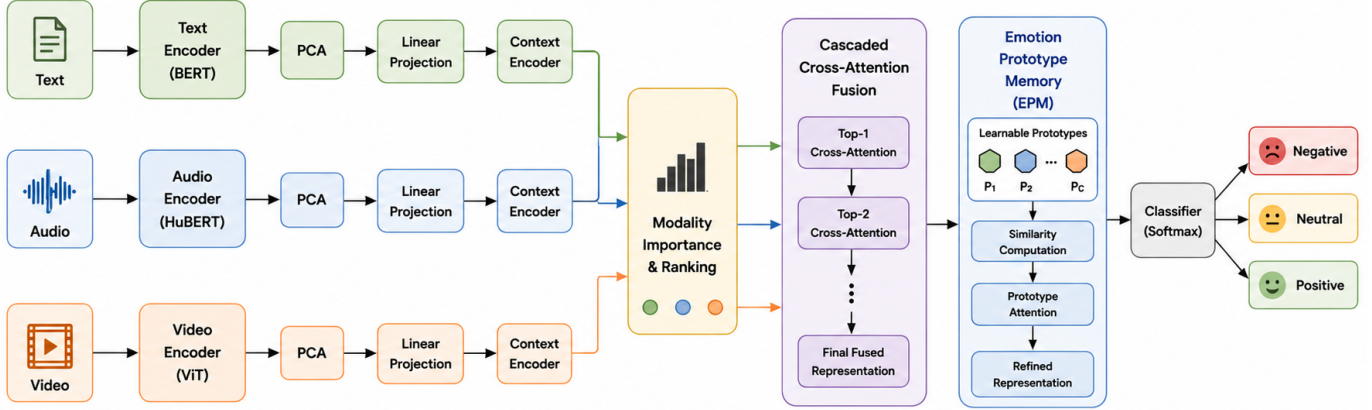


Fig. 1. Overall architecture of the proposed Emotion Prototype Memory based multimodal sentiment analysis framework.

3) *Video Feature Extraction*: Visual information provides additional emotional cues through facial expressions and appearance. In this work, visual features are extracted using the image encoder of CLIP ViT-B/32. A representative frame is selected from each video utterance and provided as input to the pretrained CLIP image encoder.

The CLIP encoder generates a high-level visual representation that captures meaningful semantic and visual characteristics of the selected frame. The extracted visual embedding has a dimension of 512.

The visual feature extraction process is defined as

$$V = \text{CLIP}(X_v), \quad (3)$$

where X_v represents the input video frame and $V \in \mathbb{R}^{512}$ denotes the extracted visual feature vector.

D. Feature Dimensionality Reduction

The feature representations extracted from BERT, HuBERT, and CLIP have different dimensions, making direct multimodal fusion challenging. In addition, high-dimensional feature vectors increase computational complexity and may contain redundant information. Therefore, Principal Component Analysis (PCA) is applied to each modality to obtain compact feature representations while preserving the most informative characteristics of the original features.

PCA transforms the original feature vectors into a lower-dimensional space by projecting them onto a set of orthogonal principal components. These components capture the maximum variance present in the data while reducing redundancy among features.

For an input feature matrix X , PCA computes the transformed representation as

$$X_{PCA} = XW, \quad (4)$$

where W represents the projection matrix formed by the selected principal components.

In the proposed framework, different target dimensions are selected for each modality according to the characteristics of the extracted features. Textual features obtained from BERT are reduced from 768 dimensions to 256 dimensions. Audio features extracted using HuBERT are reduced from 768 dimensions to 128 dimensions, while visual features extracted using CLIP are reduced from 512 dimensions to 16 dimensions. These reduced representations preserve the most relevant information while significantly decreasing the computational cost of subsequent processing stages.

The reduced feature representations are expressed as

$$T_r = \text{PCA}(T), \quad (5)$$

$$A_r = \text{PCA}(A), \quad (6)$$

$$V_r = \text{PCA}(V), \quad (7)$$

where T_r , A_r , and V_r denote the reduced text, audio, and visual feature vectors, respectively.

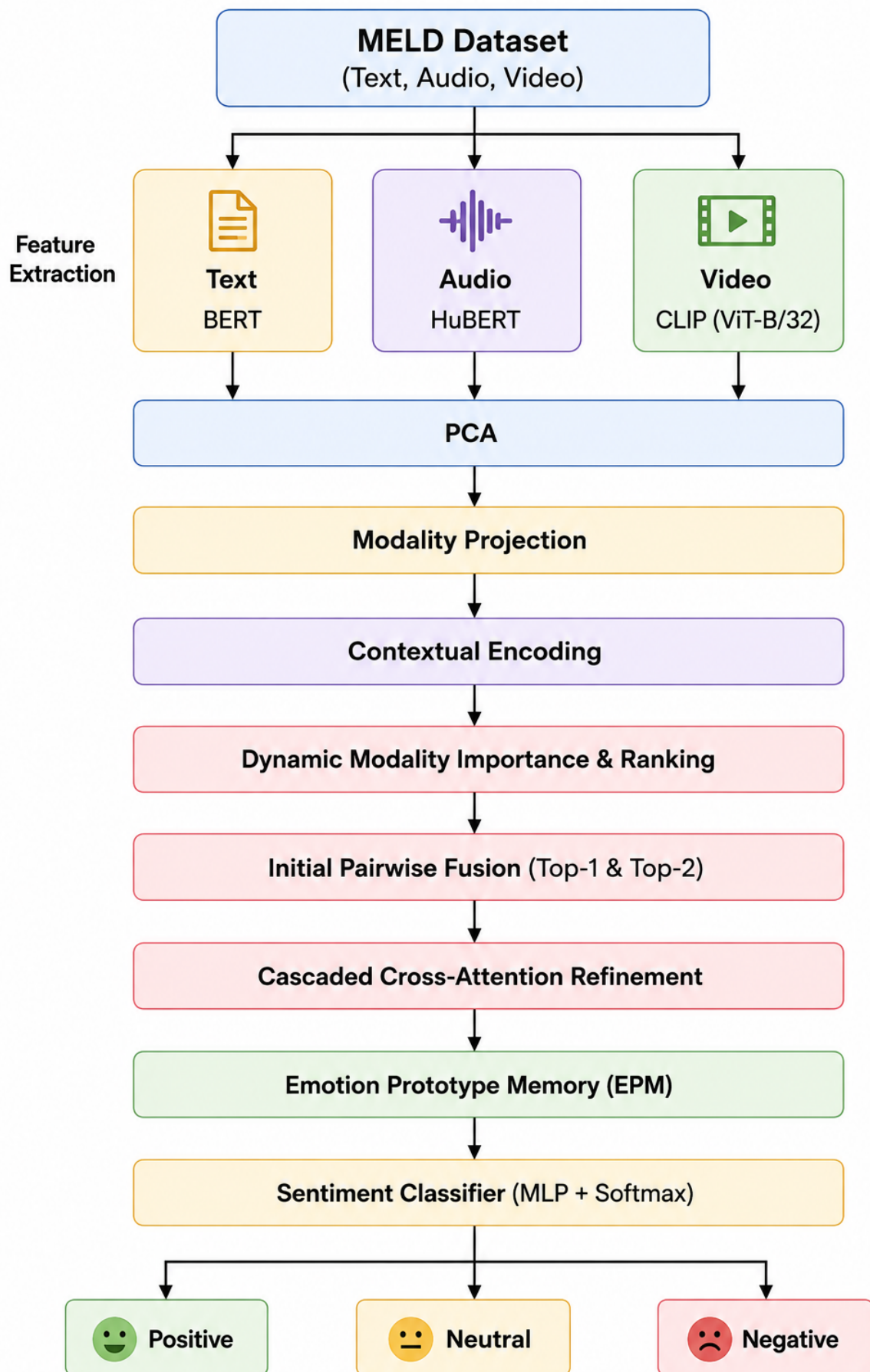


Fig. 2. Workflow of the proposed multimodal sentiment analysis framework.

TABLE I
FEATURE DIMENSIONS BEFORE AND AFTER PCA

Modality	Original Dimension	Reduced Dimension
Text (BERT)	768	256
Audio (HuBERT)	768	128
Video (CLIP ViT-B/32)	512	16

E. Modality Projection and Context Encoding

After dimensionality reduction, the feature representations of text, audio, and video have different dimensions. Since multimodal fusion requires feature vectors in a common embedding space, each modality is projected into the same feature dimension using independent linear projection layers. This transformation enables effective interaction among different modalities during the fusion process.

The projected feature representations are computed as

$$T_p = W_t T_r + b_t, \quad (8)$$

$$A_p = W_a A_r + b_a, \quad (9)$$

$$V_p = W_v V_r + b_v, \quad (10)$$

where T_r , A_r , and V_r denote the reduced feature vectors obtained after PCA. The matrices W_t , W_a , and W_v represent the learnable projection weights, while b_t , b_a , and b_v are the corresponding bias vectors.

Although the projected features lie in a common embedding space, they may still lack sufficient contextual information. Therefore, each projected feature is passed through a Transformer-based context encoder. The Transformer encoder employs a self-attention mechanism that captures contextual dependencies within each modality and generates more informative feature representations.

The context-enhanced representations are obtained as

$$T_c = \text{Transformer}(T_p), \quad (11)$$

$$A_c = \text{Transformer}(A_p), \quad (12)$$

$$V_c = \text{Transformer}(V_p), \quad (13)$$

where T_c , A_c , and V_c represent the contextual feature representations of the text, audio, and visual modalities, respectively.

The contextual representations generated by the Transformer encoder contain richer semantic information and are subsequently used by the Dynamic Modality Importance module to estimate the contribution of each modality before multimodal fusion.

F. Dynamic Modality Importance

Different modalities do not contribute equally to sentiment prediction for every input sample. For some utterances, textual information may carry the strongest sentiment cues, whereas in others, acoustic or visual information may provide more useful emotional signals. Therefore, assigning equal importance to all modalities may not result in the most effective multimodal representation.

To address this issue, the proposed framework employs a Dynamic Modality Importance (DMI) module that estimates the contribution of each modality before the fusion process. The context-enhanced representations obtained from the Transformer encoder are first passed through independent scoring layers to calculate an importance score for each modality.

The importance scores are computed as

$$s_t = f(T_c), \quad (14)$$

$$s_a = f(A_c), \quad (15)$$

$$s_v = f(V_c), \quad (16)$$

where $f(\cdot)$ represents a learnable scoring function, and s_t , s_a , and s_v denote the importance scores of the text, audio, and visual modalities, respectively.

The obtained scores are normalized using the Softmax function to generate modality importance weights:

$$\alpha_i = \frac{\exp(s_i)}{\exp(s_t) + \exp(s_a) + \exp(s_v)}, \quad (17)$$

where $i \in \{t, a, v\}$ and

$$\alpha_t + \alpha_a + \alpha_v = 1. \quad (18)$$

The normalized weights indicate the relative importance of each modality for the current input sample. A higher weight indicates that the corresponding modality provides more useful information for sentiment prediction.

The contextual feature representations are then weighted according to their importance:

$$T_w = \alpha_t T_c, \quad (19)$$

$$A_w = \alpha_a A_c, \quad (20)$$

$$V_w = \alpha_v V_c, \quad (21)$$

where T_w , A_w , and V_w represent the weighted feature representations of the text, audio, and visual modalities, respectively.

Finally, the weighted modality representations are ranked according to their importance scores before being forwarded to the cascaded cross-attention fusion module. This adaptive

weighting strategy enables the framework to focus more on informative modalities while still preserving complementary information from the remaining modalities.

G. Cascaded Cross-Attention Fusion

After estimating the importance of each modality, the weighted feature representations are fused using a cascaded cross-attention mechanism. Unlike simple concatenation, cross-attention enables one modality to attend to the informative features of another modality, thereby learning richer interactions between textual, acoustic, and visual information.

The modality with the highest importance score is selected as the primary representation, while the remaining modalities are progressively integrated according to their importance ranking. This cascaded fusion strategy allows the model to preserve the most discriminative information while gradually incorporating complementary features from the other modalities.

Given two modality representations Q and K , the cross-attention operation is computed as

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V, \quad (22)$$

where Q , K , and V denote the query, key, and value matrices, respectively, and d represents the feature dimension.

In the first stage, the highest-ranked modality attends to the second-ranked modality to generate an intermediate fused representation:

$$F_1 = \text{CrossAttention}(M_1, M_2), \quad (23)$$

where M_1 and M_2 denote the first and second ranked modalities according to their importance scores.

The intermediate representation is then combined with the remaining modality through another cross-attention operation:

$$F_2 = \text{CrossAttention}(F_1, M_3), \quad (24)$$

where M_3 represents the remaining modality. The final multimodal representation is obtained as

$$F = F_2, \quad (25)$$

where F denotes the fused feature representation containing complementary information from all three modalities.

Compared with direct feature concatenation, cascaded cross-attention allows the framework to model relationships between modalities more effectively. The adaptive fusion strategy enables the model to focus on informative modality interactions while reducing the influence of less relevant features. The resulting fused representation is subsequently refined using the proposed Emotion Prototype Memory (EPM) module before final sentiment classification.

H. Emotion Prototype Memory

Although cascaded cross-attention fusion effectively combines information from text, audio, and video, the resulting multimodal representation may still contain overlapping characteristics between different sentiment classes. This can make it difficult for the classifier to clearly distinguish between positive, neutral, and negative sentiments. To address this limitation, an Emotion Prototype Memory (EPM) module is introduced after the multimodal fusion stage.

The EPM module maintains a set of learnable prototype vectors that represent the characteristic patterns of different sentiment classes. Instead of relying only on the fused feature representation, the model compares the fused representation with these learned prototypes to retrieve additional sentiment-specific information. This enables the framework to produce a more discriminative representation before the final classification stage.

Let the fused multimodal representation obtained after cascaded cross-attention be represented as

$$F \in \mathbb{R}^d, \quad (26)$$

where d denotes the common embedding dimension.

The Emotion Prototype Memory consists of a set of learnable prototype vectors

$$P = \{P_1, P_2, \dots, P_N\}, \quad (27)$$

where N is the total number of prototypes and each prototype has the same feature dimension as the fused representation.

The similarity between the fused representation and each prototype is computed using cosine similarity:

$$S_i = \frac{F \cdot P_i}{\|F\| \|P_i\|}, \quad (28)$$

where S_i denotes the similarity score between the fused representation and the i^{th} prototype.

The similarity scores are normalized using the Softmax function to obtain attention weights:

$$\beta_i = \frac{\exp(S_i)}{\sum_{j=1}^N \exp(S_j)}. \quad (29)$$

The retrieved prototype representation is calculated as the weighted combination of all prototype vectors:

$$R = \sum_{i=1}^N \beta_i P_i, \quad (30)$$

where R represents the emotion-aware prototype representation.

The retrieved prototype information is combined with the fused multimodal representation to obtain the refined feature representation:

$$F' = F + R, \quad (31)$$

where F' denotes the refined multimodal feature vector.

The refined representation contains both the contextual information obtained from multimodal fusion and the class-specific knowledge retrieved from the Emotion Prototype Memory. This enriched representation improves the ability of the model to distinguish between similar sentiment classes and provides more informative features for the final classifier.

Since the prototype vectors are learnable parameters, they are automatically updated during the training process through backpropagation. As training progresses, the prototypes gradually learn representative characteristics of different sentiment patterns, allowing the memory module to provide more accurate guidance during sentiment prediction.

By incorporating the Emotion Prototype Memory after multimodal fusion, the proposed framework enhances the discriminative capability of the learned feature representation while preserving complementary information from all three modalities.

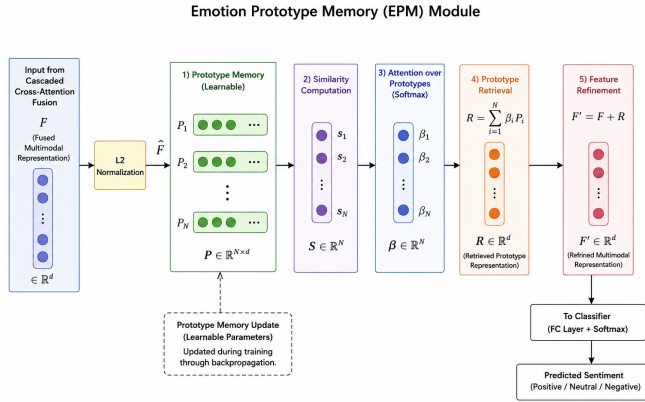


Fig. 3. Emotion Prototype Memory (EPM) module for refining the fused multimodal representation before sentiment classification.

I. Sentiment Classification

After refinement by the Emotion Prototype Memory (EPM) module, the resulting feature representation contains contextual information from all three modalities along with class-specific prototype knowledge. This refined representation is then used for the final sentiment prediction.

The refined feature vector is first passed through a fully connected layer, which learns a mapping between the multimodal representation and the sentiment classes. The output of the fully connected layer is expressed as

$$Z = W_c F' + b_c, \quad (32)$$

where F' represents the refined feature vector obtained from the EPM module, W_c and b_c denote the learnable weight matrix and bias vector of the classifier, and Z represents the output logits.

The logits are converted into class probabilities using the Softmax function:

$$P(y_i) = \frac{\exp(Z_i)}{\sum_{j=1}^C \exp(Z_j)}, \quad (33)$$

where C denotes the total number of sentiment classes and $P(y_i)$ is the probability of the i^{th} sentiment class.

The predicted sentiment label is obtained by selecting the class with the highest probability:

$$\hat{y} = \arg \max_i P(y_i). \quad (34)$$

The model is trained using the cross-entropy loss function, which measures the difference between the predicted probability distribution and the ground-truth sentiment labels. The loss function is defined as

$$\mathcal{L} = - \sum_{i=1}^C y_i \log(P(y_i)), \quad (35)$$

where y_i represents the ground-truth label and $P(y_i)$ denotes the predicted probability for the corresponding class.

During training, the parameters of all components, including the projection layers, Transformer context encoder, Dynamic Modality Importance module, cascaded cross-attention fusion module, Emotion Prototype Memory, and the final classifier, are optimized jointly through backpropagation. This end-to-end learning strategy enables the proposed framework to learn robust multimodal representations for accurate sentiment classification.

IV. EXPERIMENTAL SETUP

This section describes the dataset, preprocessing steps, implementation details, and evaluation metrics used to assess the performance of the proposed multimodal sentiment analysis framework.

A. Dataset

The proposed framework is evaluated using the Multimodal EmotionLines Dataset (MELD), a widely used benchmark for multimodal sentiment and emotion analysis. The dataset is collected from the television series *Friends* and contains multi-party conversations with synchronized textual, acoustic, and visual information.

Each utterance in the dataset is annotated with both emotion and sentiment labels. In this work, only the sentiment labels are considered, and each sample is classified into one of three categories: positive, neutral, or negative.

The official train, validation, and test splits provided with the dataset are used to ensure a fair evaluation. The dataset includes conversational utterances spoken by different speakers under diverse real-world conditions, making it suitable for evaluating multimodal sentiment analysis models.

B. Data Preprocessing

Each modality is preprocessed independently before feature extraction. Textual utterances are tokenized using the BERT tokenizer to generate contextual text representations. Audio tracks are extracted from the corresponding video clips and processed using the HuBERT feature extractor. For the visual modality, representative video frames are selected from each utterance and used as input to the CLIP image encoder.

After feature extraction, Principal Component Analysis (PCA) is applied separately to the text, audio, and visual features to reduce their dimensionality while preserving the most informative characteristics. The reduced features are then normalized and prepared for multimodal fusion.

C. Implementation Details

The proposed framework is implemented using Python and PyTorch. Pretrained BERT, HuBERT, and CLIP models are employed for extracting textual, acoustic, and visual features, respectively. Principal Component Analysis (PCA) is used for dimensionality reduction before the features are projected into a common embedding space.

The multimodal model consists of a Transformer-based context encoder, a Dynamic Modality Importance module, cascaded cross-attention fusion, and the proposed Emotion Prototype Memory (EPM). The model is trained using the Adam optimizer with cross-entropy loss for sentiment classification.

All experiments are conducted using the official training, validation, and testing splits of the MELD dataset. The same experimental settings are maintained throughout the evaluation to ensure fair performance comparison.

D. Evaluation Metrics

The performance of the proposed framework is evaluated using standard classification metrics. Accuracy is used as the primary evaluation metric to measure the percentage of correctly classified samples.

The overall accuracy is computed as

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}, \quad (36)$$

where TP , TN , FP , and FN denote the numbers of true positives, true negatives, false positives, and false negatives, respectively.

In addition to accuracy, the weighted F1-score is also considered to evaluate the balance between precision and recall across different sentiment classes. These evaluation metrics provide a comprehensive assessment of the effectiveness of the proposed multimodal sentiment analysis framework.

V. RESULTS AND DISCUSSION

This section presents the experimental results of the proposed multimodal sentiment analysis framework and evaluates the effectiveness of the proposed Emotion Prototype Memory (EPM) module. The experiments are performed on the MELD dataset using the official training, validation, and test splits.

The proposed framework is compared with the baseline multimodal model under the same experimental settings to ensure a fair comparison.

A. Performance Comparison

Table II summarizes the performance comparison between the baseline multimodal model and the proposed framework with the Emotion Prototype Memory (EPM) module. Both models use the same feature extraction, preprocessing, and training settings, with the only difference being the inclusion of the proposed EPM module.

TABLE II
PERFORMANCE COMPARISON BETWEEN THE BASELINE AND PROPOSED MODELS

Model	Test Accuracy	Weighted F1-Score
Baseline Multimodal Model	55.25%	55.64%
Proposed Model with EPM	60.92%	61.02%

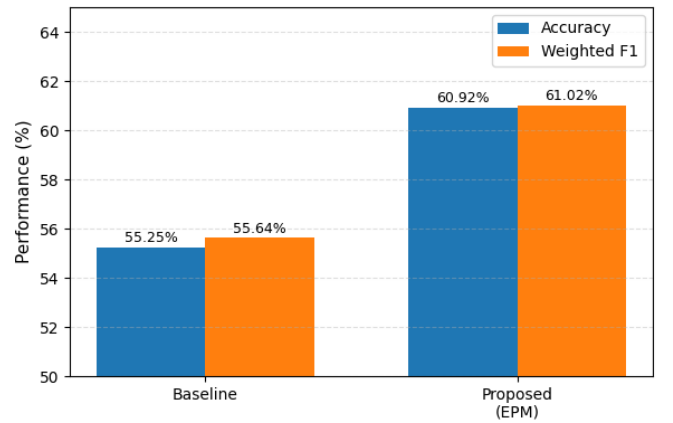


Fig. 4. Performance comparison between the baseline model and the proposed framework. The proposed model achieves higher test accuracy and weighted F1-score after incorporating the Emotion Prototype Memory (EPM).

As shown in Table II, the proposed framework achieves a test accuracy of **60.92%**, compared to **55.25%** obtained by the baseline model, resulting in an absolute improvement of **5.67%**. Similarly, the weighted F1-score increases from **55.64%** to **61.02%**. These results indicate that the proposed Emotion Prototype Memory module effectively refines the fused multimodal representation and improves the discriminative capability of the classifier for sentiment prediction.

B. Effect of Emotion Prototype Memory

The Emotion Prototype Memory (EPM) module is introduced to refine the fused multimodal representation before the final classification stage. Instead of relying only on the output of the cascaded cross-attention fusion module, EPM retrieves prototype representations that capture representative sentiment patterns learned during training.

The retrieved prototype information is combined with the fused feature representation, allowing the classifier to better distinguish between similar sentiment classes. This refinement

improves the discriminative ability of the learned features and reduces ambiguity among positive, neutral, and negative sentiments.

The experimental results indicate that the inclusion of the EPM module enhances the overall performance of the multimodal sentiment analysis framework, demonstrating the effectiveness of prototype-based feature refinement.

C. Discussion

The experimental results demonstrate that combining textual, acoustic, and visual information provides a more comprehensive understanding of sentiment than using a single modality alone. Each modality contributes complementary information, allowing the framework to capture semantic content, speech characteristics, and visual cues simultaneously.

The Dynamic Modality Importance module further improves performance by assigning adaptive importance to different modalities according to the input sample. Instead of treating all modalities equally, the model focuses more on the modality that contains stronger sentiment information while still utilizing complementary information from the remaining modalities.

The cascaded cross-attention fusion mechanism effectively models interactions between different modalities and generates richer multimodal representations. Finally, the proposed Emotion Prototype Memory module enhances these fused representations by incorporating class-specific prototype knowledge before the final classification stage. Together, these components contribute to improved sentiment prediction and demonstrate the effectiveness of the proposed multimodal framework.

VI. CONCLUSION

This paper presented a multimodal sentiment analysis framework that combines textual, acoustic, and visual information for sentiment classification. Text, audio, and video features were extracted using pretrained BERT, HuBERT, and CLIP models, respectively. After dimensionality reduction using Principal Component Analysis (PCA), the features were projected into a common embedding space and refined using a Transformer-based context encoder. The Dynamic Modality Importance module adaptively estimated the contribution of each modality, while cascaded cross-attention fusion effectively combined the multimodal representations.

To further improve sentiment classification, an Emotion Prototype Memory (EPM) module was introduced after the fusion stage. The EPM module learned representative prototype patterns and refined the fused multimodal representation before final classification. Experimental evaluation on the MELD dataset demonstrated that the proposed framework improves sentiment classification performance by generating more discriminative feature representations.

In future work, the proposed framework can be extended to support additional modalities, multilingual sentiment analysis, and missing-modality scenarios. Further improvements can also be achieved by exploring more advanced prototype

learning strategies and lightweight multimodal architectures for real-time applications.

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